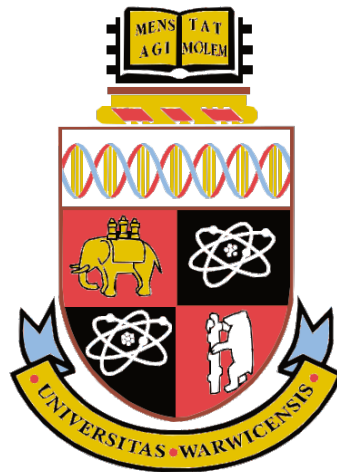




A Comparative Study of Pedestrian Dynamics Models for Evacuation Analysis

by

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Abstract

This report presents a comparative mathematical study of two approaches to pedestrian evacuation modelling: the Hughes macroscopic continuum model and the Burstedde Cellular Automata microscopic model. After surveying the landscape of pedestrian models and motivating the choice of these two specific frameworks, this report reviews how both models are constructed from physical and mathematical principles. The Hughes model couples a conservation law to an Eikonal equation for density-dependent route choice, producing a globally optimal velocity field that organises the crowd into a symmetric convergence funnel and supports sharp shock-wave solutions. The Burstedde model treats each pedestrian as a stochastic agent on a discrete grid, whose movement follows Boltzmann-type transition probabilities derived from static and dynamic floor fields.

Both models are implemented in Python and applied to the evacuation of a 10 m × 10 m square room through a single 1 m exit centred on the top wall. The Hughes simulation is studied in two spatial dimensions, where the coupling between the conservation law and the Eikonal equation introduces non-trivial implementation challenges including singularity avoidance and numerical stability verification. The CA simulation of 800 agents across 10 realisations reveals an emergent triangular depletion pattern and significant stochastic variability across realisations. The two models are compared systematically across mathematical character, computational complexity, physical realism, and practical implementation difficulty. The Hughes model's globally optimal route choice and analytical tractability are offset by its inability to reproduce individual-level phenomena; the CA model's emergent bottleneck dynamics and stochastic realism come at the cost of greater computational burden and harder parameter calibration. Hybrid multiscale approaches are identified as the most promising direction for future work.

For this project, I wrote Python code for both models. This code can be found on my GitHub page via the following link: <https://github.com/TristanHartwell403/MA4K8-Project-Tools>

The code can be manipulated for different scenarios with different input variables and isn't necessarily in the exact form to produce all figures in this report.

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1 Introduction

1.1 Motivation

Understanding how large groups of people move through confined spaces is one of the most practically consequential problems in applied mathematics. Crowd-related disasters occur with alarming regularity: the 2021 Astroworld Festival crush in Houston resulted in ten deaths; the 2010 Love Parade disaster in Duisburg killed twenty-one people; the 2015 Mina stampede during the Hajj pilgrimage caused over two thousand fatalities. In each case, a better understanding of crowd flow dynamics could have informed the design of safer spaces, clearer exit strategies, or more effective crowd management protocols.

Mathematical modelling provides a systematic framework for studying such phenomena without requiring dangerous real-world experiments. By simulating how crowd density or individual agents evolve under prescribed rules, we can probe questions that empirical observation alone cannot easily answer. How does a small change in exit width affect evacuation time? At what crowd density does a bottleneck become critical? Does routing pedestrians more aggressively toward an exit actually help, or can it make things worse?

1.2 Classes of Pedestrian Dynamics Model

Mathematical models for pedestrian dynamics form a hierarchy defined by the scale at which individual behaviour is resolved. This section surveys the principal classes and explains the motivations behind each, before justifying the specific choices made in this report.

1.2.1 Macroscopic Models

Macroscopic, or continuum, models treat the crowd as a compressible fluid. Rather than tracking any individual pedestrian, they describe the crowd through a density field $\rho(\mathbf{x}, t)$ (people per square metre) that evolves according to partial differential equations. This approach draws directly on the analogy between pedestrian flow and vehicular traffic identified independently by Henderson [10] and formalised rigorously in the Lighthill-Whitham-Richards (LWR) theory [15]: a crowd in a corridor behaves much like a column of vehicles, with slow movers at the front creating a backward-propagating density wave.

The central strength of macroscopic models is their analytical and computational tractability. For a domain discretised on an $N \times N$ grid the cost per time step is $\mathcal{O}(N^2)$, regardless of the number of people in the crowd. The tools of conservation law theory - characteristics, Rankine-Hugoniot conditions, entropy solutions - provide rigorous, quantitative predictions about shockwave speeds, density profiles, and evacuation times. In one spatial dimension these predictions are exact; in two dimensions they extend to the qualitative description of congestion fronts and rarefaction waves. For large crowds the continuum

approximation is well-justified: with hundreds or thousands of pedestrians in a room, individual fluctuations average out smoothly over the scales of interest [10].

Important examples of macroscopic models include the pure LWR model [15]; the Payne-Whitham model [4], which adds inertia terms analogous to compressible gas dynamics; and the Hughes model [11], which adds a density-dependent route-choice mechanism via the Eikonal equation.

The central limitation of macroscopic models is the loss of individual information. A density field cannot represent heterogeneous populations (for example the elderly or mobility-impaired individuals), track whether any specific person escaped, or reproduce phenomena that depend on individual interactions. These include: *arching at bottlenecks* (a stable high-density arch that forms just behind a narrow exit, caused by mutual blocking - pedestrians competing to enter the gap block each other and a curved queue builds up); *lane formation* (spontaneous segregation into opposing directional streams in bidirectional flows, reducing mutual interference); *herding* (the tendency to follow paths recently used by others, producing collective flow trails); and the *faster-is-slower effect* (increasing individual urgency paradoxically slows evacuation by intensifying conflicts at the bottleneck) [8]. These phenomena arise from individual interactions and are inherently absent from smooth density-field descriptions.

1.2.2 Microscopic Models

Microscopic, or agent-based, models sit at the opposite extreme of the resolution spectrum. Each pedestrian is represented as a discrete entity with their own position, velocity, and decision-making mechanism, and collective behaviour emerges from the simultaneous application of simple local rules to all agents. The paradigmatic example is the Social Force Model introduced by Helbing and Molnár [9], in which each pedestrian obeys Newton’s second law subject to a driving force toward the exit, repulsive forces from nearby pedestrians and walls, and attractive forces from companions. The resulting system of ordinary differential equations is integrated numerically at cost $\mathcal{O}(N_p^2)$ per step (or $\mathcal{O}(N_p)$ with spatial hashing).

The intellectual justification for microscopic models is that many important crowd phenomena are inherently individual. Lane formation in bidirectional flows, clogging at bottlenecks, and oscillating flow patterns near narrow passages all emerge naturally from local repulsion rules and have no counterpart in purely macroscopic descriptions.

The cellular automata (CA) approach provides a computationally simpler alternative to the social force model. Rather than treating space as continuous, the CA model discretises it into a grid of cells each occupied by at most one agent. Movement is stochastic: at each time step each agent independently draws a random move from a probability distribution over neighbouring cells. The CA framework was applied to pedestrian dynamics by

Burstedde et al. [2] and has since become one of the most widely studied microscopic approaches, with its discrete character making it particularly tractable for large-scale Monte Carlo studies and extensions to heterogeneous populations [16].

The cost of microscopic resolution is computational: each time step requires $\mathcal{O}(N_p)$ work for N_p agents, and the stochastic nature of CA models means that reliable statistical conclusions require many independent simulation runs. Furthermore, the parameters of microscopic models are often difficult to measure directly and must be inferred from observed crowd behaviour [18].

1.2.3 Mesoscopic Models

Mesoscopic models occupy a middle ground, describing the crowd through a probability distribution over individual states rather than a density field or explicit agents. The kinetic theory approach of Bellomo and Dogbe [1] models the crowd through a distribution function $f(\mathbf{x}, \mathbf{v}, t)$ over position and velocity space, satisfying a Boltzmann-type integro-differential equation. This captures more individual-level detail than macroscopic models while remaining more analytically tractable than full agent simulations. We mention this class for completeness but do not study it further here.

1.2.4 Choice of Geometry, Dimensionality, and Models

Why two spatial dimensions? Both models in this study are implemented and studied in two spatial dimensions. This is a deliberate and significant choice. In one dimension - a straight corridor with a single exit at one end - the Hughes model reduces exactly to the Lighthill-Whitham-Richards traffic equation, for which the full analytical theory of entropy solutions, characteristics, and shocks is completely established. The two-dimensional setting is substantially more challenging: the Eikonal equation now governs a two-dimensional potential field that must route the crowd around arbitrary obstacles and through a finite-width exit, producing curved streamlines, non-trivial iso-cost contours, and the potential for the turning-point singularity to manifest in multiple directions simultaneously. The shockwave structure is also richer: in 2D, density fronts are curves rather than points, and their propagation speed varies with local density and geometry. Studying these phenomena in two dimensions is not only more realistic for actual evacuation scenarios (rooms, not corridors) but also mathematically more interesting, and the numerical implementation challenges - stable upwind schemes in 2D, convergence of the fast sweeping solver, gradient computation near walls - are substantially harder than their 1D counterparts.

Two dimensions is also sufficient to capture realistic evacuation dynamics in buildings. Although real buildings are three-dimensional, humans do not truly move through three-dimensional space during an evacuation: movement is almost always confined to horizontal floor plates. Vertical transitions via stairs, ramps, and escalators are conventionally modelled as a network of expanded two-dimensional floor spaces connected at nodes, so

three-dimensional buildings are typically handled by extending two-dimensional models to a multi-floor network rather than replacing them with a three-dimensional PDE or agent model [16]. Two dimensions therefore provides the natural and sufficient level of description for the pedestrian dynamics studied here.

Why a square room with a single exit? The square room with a single exit is the standard benchmark geometry in the pedestrian dynamics literature [2, 11]. Its simplicity makes it ideal for comparing models: any qualitative differences between the Hughes and CA outputs arise from the model structure rather than geometric complexity. The single exit creates a genuine bottleneck, allowing the arch formation, congestion wave, and evacuation rate to be clearly observed and compared. More complex geometries (multiple exits, corridors, L-shaped rooms) were explored briefly in the early stages of this work and produced qualitatively similar phenomena with additional complexity that obscured the model comparison; the square room was therefore retained as the primary scenario. The centred exit was chosen (rather than a corner exit) because it produces the symmetric convergence funnel that most clearly illustrates the global route-choice mechanism of the Hughes model.

Why the Hughes and Burstedde models? With the two-dimensional square-room setting established, this report studies the Hughes macroscopic model [11] and the Burstedde CA microscopic model [2]. These two were chosen for four reasons. First, they represent the most clearly contrasting approaches in the literature: one is deterministic, continuous, and analytically rich; the other is stochastic, discrete, and computationally direct. This contrast makes them ideal for a comparative study. Second, both are well-established: the Hughes model has been studied rigorously by Di Francesco et al. [5] and extended by numerous authors; the Burstedde model has been implemented, validated, and extended in hundreds of published studies. Third, both are practically implementable in Python without specialist libraries, enabling a direct comparison of implementation difficulty as well as mathematical properties. Fourth, the two models are formally connected through the hydrodynamic limit, a theoretical link discussed in Section 4.1 that makes the comparison especially illuminating.

1.3 Project Aims and Scope

The aims of this report are:

1. To review and present the mathematical construction of both models, tracing their governing equations from physical principles and analysing their key mathematical properties in two spatial dimensions.
2. To implement both models numerically in Python, describing discretisation schemes, solvers, and algorithmic choices in sufficient detail for reproducibility.

3. To simulate the evacuation of a $10\text{ m} \times 10\text{ m}$ square room through a single 1 m exit centred on the top wall, and present density fields, evacuation curves, and spatial flow patterns.
4. To compare the two models across mathematical character, computational cost, physical realism, and implementation difficulty.
5. To draw practical conclusions about which model is better suited to which class of application.

Section 2 develops the Hughes model; Section 3 develops the CA model; Section 4 provides a comprehensive comparative analysis; Section 5 summarises and discusses future directions.

2 The Hughes Macroscopic Model

2.1 Derivation of the Model

2.1.1 The Crowd as a Continuum

Hughes [11] proposed treating a large crowd as a compressible fluid, building directly on the LWR traffic analogy. The fundamental object is the pedestrian density $\rho(\mathbf{x}, t)$ at position $\mathbf{x} \in \Omega \subset \mathbb{R}^2$ and time $t \geq 0$. The continuum approximation is valid when individual fluctuations average out over the scales of interest. Henderson [10] provided early empirical support for this approximation, showing that pedestrian density distributions obey statistical laws analogous to those of classical fluids. In a $10\text{ m} \times 10\text{ m}$ room at around 4 people/m^2 we assume there are 384 people (slightly less than 400 as the walls have non-zero thickness); the mean inter-person spacing of around 0.5 m is larger than our typical iterative distance step in the simulation with spacing of 0.1 m , so the density field should be understood as a coarse-grained or smoothed representation of the true discrete crowd.

2.1.2 Conservation of Mass

In the absence of sources or sinks, the density satisfies the continuity equation obtained by applying the divergence theorem to conservation of mass in an arbitrary sub-region:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0, \quad (1)$$

where $\mathbf{v}(\mathbf{x}, t)$ is the macroscopic velocity field. A large amount of the modelling content of the Hughes model resides in the choice of $\mathbf{v}$.

2.1.3 Velocity and Route Choice via the Eikonal Equation

Hughes' key contribution was to model pedestrians as rational agents who simultaneously walk at a speed determined by local density and choose routes minimising total travel cost

to the exit, accounting for both distance and congestion. Let $U(\rho)$ denote walking speed at density ρ , and $G(\rho)$ a discomfort weight encoding psychological cost per unit distance. An agent minimising total cost follows the direction of steepest descent of a potential $\phi(\mathbf{x}, t)$ satisfying the Eikonal equation:

$$|\nabla\phi| = \frac{G(\rho)}{U(\rho)} =: f(\rho), \quad \phi|_{\partial\Omega_{\text{exit}}} = 0. \quad (2)$$

The velocity field is:

$$\mathbf{v} = -U(\rho) \frac{\nabla\phi}{|\nabla\phi|}. \quad (3)$$

The coupled system (1)-(3) is the full Hughes model. The boundary condition $\phi = 0$ at the exit reflects that the perceived cost of reaching the exit from the exit is zero; ϕ increases in all directions away from the exit, with the rate of increase determined by $f(\rho)$.

2.2 The Speed and Discomfort Functions

2.2.1 Speed Function, Parameter Choices, and Singularity Avoidance

We use the linear (Greenshields-type) speed-density relationship [7]:

$$U(\rho) = v_{\max} \left(1 - \frac{\rho}{\rho_{\max}} \right), \quad (4)$$

with $v_{\max} = 1.4 \text{ m/s}$ and $\rho_{\max} = 8 \text{ people/m}^2$. The free-flow speed $v_{\max} = 1.4 \text{ m/s}$ is consistent with Weidmann's [19] widely-cited empirical study, which reports a mean unobstructed walking speed of 1.34 m/s (standard deviation 0.26 m/s). The jam density $\rho_{\max} = 8 \text{ people/m}^2$ is a physically reasonable central estimate; empirical studies [6, 18] report jam densities in the range $6\text{-}10 \text{ people/m}^2$.

Implementation: the speed floor. If $U(\rho) \rightarrow 0$ as $\rho \rightarrow \rho_{\max}$, the Eikonal cost $f(\rho) = G(\rho)/U(\rho) \rightarrow \infty$, causing numerical instability in the Fast Sweeping Method (FSM) solver (discussed in detail in Section 2.5). To prevent this, a floor is imposed on the speed function in the implementation:

$$U_{\epsilon}(\rho) = v_{\max} \max\left(\epsilon, 1 - \frac{\rho}{\rho_{\max}}\right), \quad \epsilon = 10^{-3}, \quad (5)$$

ensuring $U(\rho) \geq \epsilon v_{\max} > 0$ everywhere. The theoretical motivation for this regularisation is discussed in Section 2.3.2.

2.2.2 Discomfort Function and the Role of α

The discomfort function encodes the preference for less-congested routes. Following Hughes [11] we use:

$$G(\rho) = \frac{1}{1 + \alpha\rho}, \quad \alpha = 0.1, \quad (6)$$

decreasing from $G(0) = 1$ to $G(\rho_{\max}) \approx 0.56$ at jam density. If $G \equiv 1$ (i.e. $\alpha = 0$), pedestrians minimise Euclidean distance to the exit regardless of congestion; for $\alpha > 0$ they route around dense areas, accepting longer paths to avoid the crowd.

The role of α in the existence of solutions. The parameter α has a subtle effect on well-posedness. When $\alpha = 0$, the Eikonal equation decouples from ρ (since $f(\rho) = 1/U(\rho)$ depends on ρ only through U), and the turning-point analysis of Di Francesco et al. [5] applies in its standard form. When $\alpha > 0$, the additional ρ -dependence in $G(\rho)$ makes the coupling stronger: Eikonal iso-cost contours distort more sharply around density concentrations, and no exact general solution is known. The value $\alpha = 0.1$ follows Hughes' original paper [11] and was found to produce physically sensible, stable solutions; stability degrades as α is increased significantly beyond 1, consistent with stronger nonlinearity in the coupling.

ρ (people/m ²)	$U(\rho)$ (m/s)	$G(\rho)$	$f(\rho) = G/U$	Regime
0.0	1.400	1.000	0.714	Free flow
2.0	1.050	0.833	0.794	Light crowd
4.0	0.700	0.714	1.020	Moderate (initial)
6.0	0.350	0.625	1.786	Dense
8.0	0.001	0.556	556.0	Near jam (regularised)

Table 1: Values of $U(\rho)$, $G(\rho)$, and $f(\rho) = G(\rho)/U(\rho)$ evaluated at five representative densities ρ (parameters: $v_{\max} = 1.4$ m/s, $\rho_{\max} = 8$ people/m², $\alpha = 0.1$). Each row gives the value of these functions at the specific ρ shown in the first column. The steep rise in $f(\rho)$ near jam density drives dynamic routing: congested cells have high perceived cost, so the Eikonal field deflects pedestrians around them.

Gradient singularity of ϕ . Even with $U_\epsilon > 0$, there can be isolated cells where $|\nabla\phi|$ is numerically very small (near exit cells where $\phi = 0$). Division by a near-zero $|\nabla\phi|$ would produce spuriously large velocities, so $|\nabla\phi|$ is floored at 10^{-8} before division. More fundamentally, at the turning-point density $\rho^* = \rho_{\max}/2$, the characteristic speed $c(\rho^*) = 0$ and Di Francesco et al. [5] show that $\nabla\phi$ can become ill-defined, making the velocity field (3) undefined without regularisation. Since the initial density $\rho_0 = 4.0$ people/m² = $\rho_{\max}/2$ lies exactly at this critical value, the $|\nabla\phi|$ floor is relevant from the very first time step.

2.3 Mathematical Properties

2.3.1 Connection to Traffic Flow Theory and the Fundamental Diagram

Although this report focuses on the two-dimensional model, it is useful to note that in a one-dimensional corridor the Hughes system reduces to the Lighthill-Whitham-Richards traffic model [15] with flux $q(\rho) = \rho U(\rho) = v_{\max} \rho (1 - \rho/\rho_{\max})$. This connection confirms the physical soundness of the Hughes model and provides the vocabulary - free-flow, congested, shockwave, rarefaction - used to describe the 2D dynamics. In two dimensions the Eikonal equation is non-trivially coupled to the density field and the problem is substantially harder, as discussed below.

The characteristic speed of density information is $c(\rho) = q'(\rho) = v_{\max}(1 - 2\rho/\rho_{\max})$. When faster information (from low-density regions) overtakes slower information (from high-density regions), a shockwave forms. The shock speed satisfies the Rankine-Hugoniot condition:

$$s = \frac{q(\rho_R) - q(\rho_L)}{\rho_R - \rho_L} = v_{\max} \left(1 - \frac{\rho_L + \rho_R}{\rho_{\max}} \right), \quad (7)$$

and the Lax entropy condition $c(\rho_L) > s > c(\rho_R)$ selects the physically admissible shock. These results apply in 2D wherever the density front is locally one-dimensional (perpendicular to the flow direction), as is approximately the case for the downward-propagating density front observed in our simulations.

2.3.2 The Turning Point and Non-Existence of Classical Solutions

A crucial distinction between the Hughes model and the pure LWR model is the coupling to the Eikonal equation at every time step. Di Francesco et al. [5] showed rigorously that this coupling introduces a critical density:

$$\rho^* = \frac{\rho_{\max}}{2}, \quad (8)$$

at which $c(\rho^*) = 0$. At this *turning point*, the potential ϕ may develop a singularity: the velocity field (3) becomes undefined, and classical solutions do not generally exist. The $|\nabla\phi|$ floor and the speed regularisation (5) together prevent this singularity from existing: the first stops $|\nabla\phi| \rightarrow 0$ at the turning point and the second stops $U \rightarrow 0$ near jam density. The existence and uniqueness of entropy solutions for the 2D coupled system remains an open problem; Di Francesco et al. [5] resolve it in 1D but the 2D case requires new techniques.

2.4 Numerical Implementation

2.4.1 Domain and Grid

The square room $[0, 10]^2 \text{ m}^2$ is discretised on a 100×100 uniform grid with $\Delta x = \Delta y = 0.1 \text{ m}$. Grid point (i, j) represents position $(i\Delta x, j\Delta y)$. Boundary cells form the walls and are permanently blocked with $\rho = 0$. The exit is a 1 m gap (10 cells) centred on the top wall ($y = 10 \text{ m}$), corresponding to columns $j \in [45, 54]$ at row $i = 99$.

2.4.2 Upwind Finite Difference Scheme

The continuity equation (1) is advanced using a first-order explicit upwind scheme. The upwind approach is essential for hyperbolic PDEs: it respects the directionality of information propagation, using upstream data to compute fluxes and thereby preventing the non-physical oscillations (Gibbs phenomenon) that central-difference schemes generate near shockwaves. The update at interior cell (i, j) is:

$$\rho_{i,j}^{n+1} = \rho_{i,j}^n - \Delta t \left[\frac{F_{i+1/2,j}^n - F_{i-1/2,j}^n}{\Delta x} + \frac{G_{i,j+1/2}^n - G_{i,j-1/2}^n}{\Delta y} \right], \quad (9)$$

where the interface fluxes are:

$$F_{i+1/2,j} = \max(v_{x,i,j}, 0)\rho_{i,j} + \min(v_{x,i+1,j}, 0)\rho_{i+1,j}, \quad (10)$$

$$G_{i,j+1/2} = \max(v_{y,i,j}, 0)\rho_{i,j} + \min(v_{y,i,j+1}, 0)\rho_{i,j+1}. \quad (11)$$

These formulas select the upwind value of ρ : when flow is in the positive direction ($v > 0$), the flux uses the left/lower cell; when flow is in the negative direction, it uses the right/upper cell. The scheme is first-order accurate with truncation error $\mathcal{O}(\Delta x) + \mathcal{O}(\Delta t)$; the leading error term is a numerical diffusion proportional to $v_{\max} \Delta x \partial^2 \rho / \partial x^2$, which slightly smears shockwaves over $\mathcal{O}(1/\Delta x)$ cells. Higher-order schemes such as MUSCL or ENO/WENO [17] would reduce this smearing but are significantly more complex to implement.

2.4.3 The CFL Condition

Stability requires that the numerical domain of dependence contain the analytical domain of dependence of the PDE. For the 2D continuity equation with maximum wave speed $v_{\max}$, this yields the Courant-Friedrichs-Lewy (CFL) condition:

$$\text{CFL} = v_{\max} \Delta t \left(\frac{1}{\Delta x} + \frac{1}{\Delta y} \right) \leq 1. \quad (12)$$

With $v_{\max} = 1.4 \text{ m/s}$ and $\Delta x = \Delta y = 0.1 \text{ m}$, this gives $\Delta t \leq 1/(2 \times 1.4 \times 10) \approx 0.036 \text{ s}$. We use $\Delta t = 0.05 \text{ s}$, satisfying CFL with a safety margin. If CFL is violated, small numerical

errors are amplified exponentially at each step, rapidly destroying the solution.

2.4.4 Fast Sweeping for the Eikonal Equation

At each time step the Eikonal equation $|\nabla\phi| = f(\rho)$ must be solved on the current density field to determine the velocity for the next step. We use the Fast Sweeping Method (FSM) of Zhao [20]. The FSM initialises $\phi = +\infty$ everywhere except at exit cells ($\phi = 0$) and performs four Gauss-Seidel sweeps of the grid in alternating diagonal directions. At each interior cell (i, j) the local quadratic

$$\frac{(\phi_{i,j} - \phi_x)_+^2}{\Delta x^2} + \frac{(\phi_{i,j} - \phi_y)_+^2}{\Delta y^2} = f(\rho_{i,j})^2, \quad (13)$$

is solved, where $\phi_x = \min(\phi_{i-1,j}, \phi_{i+1,j})$ and $\phi_y = \min(\phi_{i,j-1}, \phi_{i,j+1})$ are the smallest values in each coordinate direction, and $(x)_+ = \max(x, 0)$. When the two upwind values differ by more than $f(\rho_{i,j}) \min(\Delta x, \Delta y)$, a simpler one-dimensional update $\phi_{i,j} = \min(\phi_x, \phi_y) + f \cdot \min(\Delta x, \Delta y)$ is used; otherwise the full quadratic is solved. The cell value is updated only if the new value is smaller, ensuring the FSM is monotone. For our convex square domain with a single exit, convergence to tolerance 10^{-6} occurs within 3 to 4 sweeping passes [20]. The complexity per time step is $\mathcal{O}(N^2)$, slightly better than the Fast Marching Method's $\mathcal{O}(N^2 \log N)$ [17] due to the absence of heap operations.

2.4.5 Velocity and Boundary Conditions

Given ϕ , the gradient $\nabla\phi$ is computed using central differences in the interior and one-sided differences adjacent to walls, with $|\nabla\phi|$ floored at 10^{-8} to avoid division by zero. Wall cells enforce no-penetration by being excluded from the region over which the FSM sweeps are applied: the solver simply never updates ϕ at wall cells, so no velocity is ever computed there (an alternative approach of assigning $\phi = +\infty$ to wall cells was tested but produced large, spurious potential gradients near the boundaries that destabilised the solver; excluding wall cells from the sweep domain proved more robust). Exit cells are absorbing: density flowing through the exit is removed at each time step, representing pedestrians leaving the room.

2.5 Implementation Challenges and Numerical Validation

Before presenting simulation results, we document the key implementation challenges encountered and the validation steps taken to ensure the numerical solution is reliable. These are mathematically interesting in their own right and are not standard in textbook PDE solvers.

2.5.1 Singularities in the Coupled PDE System

The main source of numerical difficulty in the Hughes model is the tight coupling between the density update and the Eikonal solve at every time step. Three distinct singularity issues arise.

Speed function singularity. As noted in Section 2.2, $U(\rho) \rightarrow 0$ as $\rho \rightarrow \rho_{\max}$, causing $f(\rho) = G(\rho)/U(\rho) \rightarrow \infty$. In the FSM, this produces ϕ values that grow without bound in near-jam cells, leading to enormous velocity magnitudes and immediate instability. The floor (5) with $\epsilon = 10^{-3}$ caps $f(\rho)$ at $G(8)/(10^{-3} \times 1.4) \approx 397$, keeping the Eikonal solve well-conditioned throughout.

Gradient singularity. Even when U is bounded away from zero, the velocity $\mathbf{v} = -U(\rho)\nabla\phi/|\nabla\phi|$ requires dividing by $|\nabla\phi|$. Near isolated cells where the potential surface is locally flat (near $\phi = 0$ at exit cells), $|\nabla\phi|$ can be numerically zero. A floor of 10^{-8} is applied to $|\nabla\phi|$ before division.

Turning-point singularity. At $\rho = \rho^* = \rho_{\max}/2 = 4 \text{ people/m}^2$, the characteristic speed $c(\rho^*) = 0$ and the coupled PDE system does not admit classical solutions in general [5]. The $|\nabla\phi|$ floor of 10^{-8} prevents the turning-point singularity from manifesting by ensuring the velocity field remains well-defined even when $\nabla\phi$ becomes very small. The initial density $\rho_0 = 4.0 \text{ people/m}^2 = \rho_{\max}/2$ was chosen as a physically realistic moderate crowd density. It happens to coincide with the turning-point density ρ^* , which makes the gradient regularisation relevant from the outset.

2.5.2 Stability Verification Across Grid Sizes and Time Steps

A critical question for any numerical PDE simulation is whether the computed solution is a faithful approximation to the true solution, or whether it is an artefact of the particular discretisation chosen. To verify stability and convergence, we checked the following:

CFL stability. The time step $\Delta t = 0.05 \text{ s}$ satisfies the CFL condition (12) with a safety factor of approximately 0.7. We verified that halving Δt to 0.025 s (tightening the bound) produces density snapshots and evacuation curves that are visually indistinguishable from those at $\Delta t = 0.05 \text{ s}$, confirming that $\Delta t = 0.05 \text{ s}$ is already in the stable regime and that the solution is not sensitive to the precise time step within the stable range.

Grid convergence. Reducing the grid from 100×100 to 50×50 (doubling Δx and Δy to 0.2 m) produces the same three-phase structure (funnel, shockwave, arch) with the shockwave propagating at the same speed, though with visibly more numerical diffusion (blurring of the shock front, consistent with first-order accuracy). This confirms that $\Delta x = 0.1 \text{ m}$ provides sufficient resolution to capture the essential physics, while coarser grids preserve qualitative behaviour at reduced computational cost. The chosen values $\Delta x = 0.1 \text{ m}$ and $\Delta t = 0.05 \text{ s}$ therefore represent a pragmatic balance between numerical precision

and computational feasibility; refining further would improve quantitative accuracy but not change the qualitative picture.

FSM convergence. The FSM is run for up to 6 sweep iterations per time step, with early termination when the change in ϕ between iterations is below 10^{-6} . In practice, convergence occurs within 3-4 iterations throughout the simulation, confirming that the ϕ field is well-resolved at every time step.

2.6 Simulation Results: Square Room, Top-Middle Exit

We focus on the early stages of the evacuation because these are the most critical and interesting, so we model the first 60s of the evacuation. Table 2 summarises the simulation parameters. The run completed in 775 s for 1,200 time steps covering the 60 s of simulated time, with an initial population of 384.2 people.

Parameter	Value	Units	Source/justification
$v_{\max}$	1.4	m/s	Weidmann [19]
$\rho_{\max}$	8.0	people/m ²	Fruin [6], Seyfried et al. [18]
ρ_0	4.0	people/m ²	Maximum-throughput density ($\rho_{\max}/2$)
α	0.1	m ² /person	Hughes [11]
ϵ	10^{-3}	-	Numerical regularisation
$N_x = N_y$	100	cells	Grid resolution
$\Delta x = \Delta y$	0.1	m	Grid spacing
Δt	0.05	s	CFL constraint
Exit width	1.0	m (10 cells)	Centred on top wall

Table 2: Hughes model simulation parameters with justifications.

2.6.1 The Eikonal Field at Uniform Density

Figure 1 shows the Eikonal potential ϕ computed from a uniform unit cost field ($f \equiv 1$, corresponding to zero density), giving the distance-to-exit field in the absence of crowd effects. The field decreases symmetrically from the bottom corners toward the exit at the top centre, with smooth contours fanning outward from the exit point. This baseline illustrates an important property: the Eikonal equation routes the crowd globally and symmetrically even before any density effects develop, creating a funnelling geometry that a purely local rule cannot replicate.

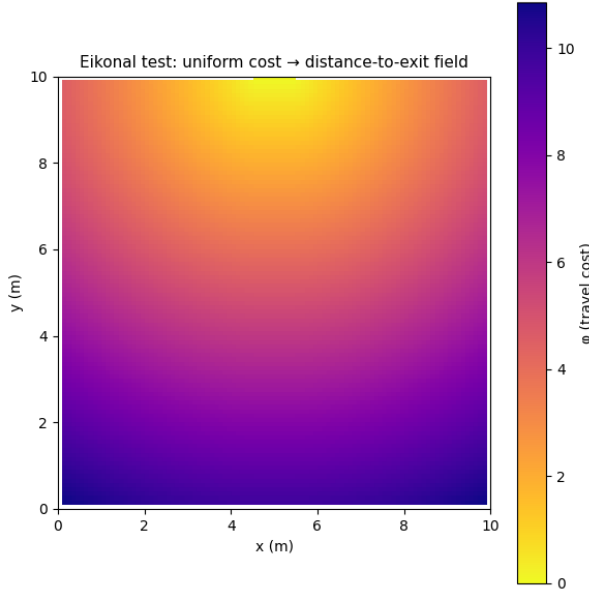


Figure 1: Eikonal potential ϕ from a uniform cost field ($f \equiv 1$). Iso-potential contours fan symmetrically from the top-centre exit, representing equal-cost-to-exit curves. At $\rho = 0$ the field is the Euclidean distance to the exit; at positive density the contours distort to reflect the higher cost of traversing dense regions. This figure is the author's own output from the Python implementation.

2.6.2 Density Evolution and Phase Structure

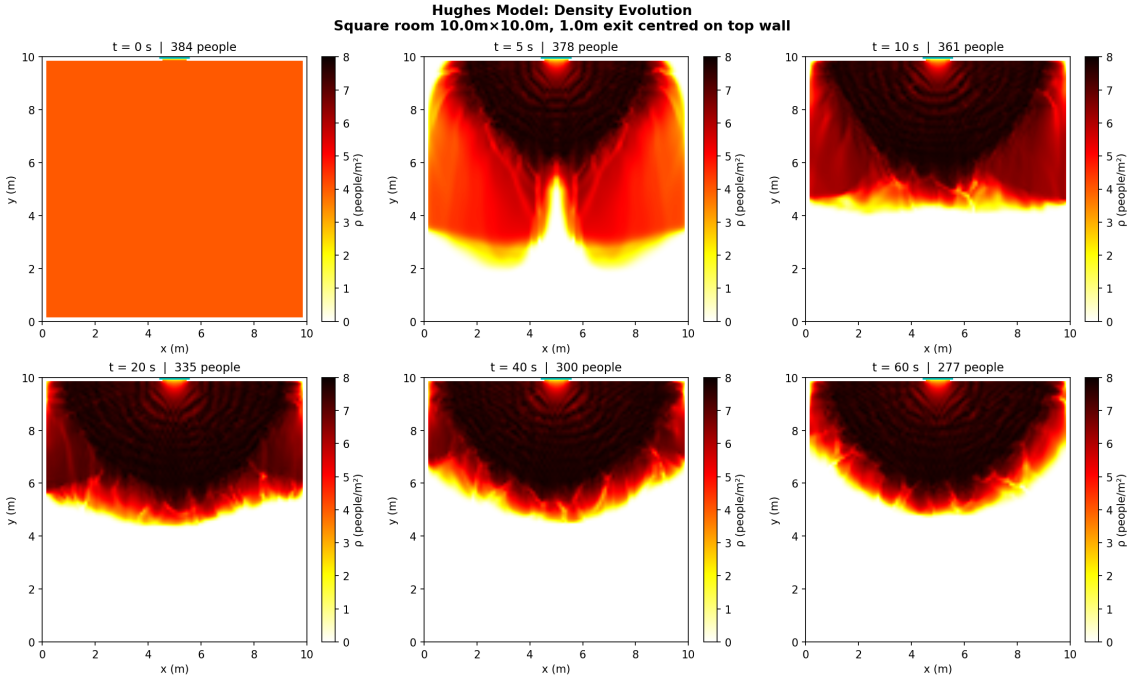


Figure 2: Hughes model density evolution at $t = 0, 5, 10, 20, 40, 60$ s. Colour scale: yellow/light = low density, dark red/black = near jam density (8 people/m^2). All figures from this model are the author's own outputs. Cyan bar marks the exit.

Figure 2 shows the density field at six times. The evolution exhibits three clearly distinguishable phases. At $t = 5$ s a low-density funnel channel has opened up along the path to the exit, flanked by higher-density lobes on either side - the crowd has self-organised

into an optimal flow pattern. By $t = 10$ s the shockwave front and concentric Eikonal contours are clearly established; at $t = 20$ - 60 s a persistent semicircular high-density arch dominates.

Phase 1 - Symmetric convergence ($t = 0$ to 5 s). At $t = 0$ the room is uniformly filled at $\rho_0 = 4.0$ people/m². By $t = 5$ s a striking symmetric structure has formed: a low-density funnel-shaped channel has opened up along the direct path to the exit (people have already flowed through this corridor toward the exit, depleting the density there), flanked on either side by two higher-density lobes - the regions of the room that the Eikonal routing has not yet directed toward the exit. The crowd organises itself into this optimal flow pattern entirely through the global Eikonal computation; no counterpart exists in local models such as the CA.

Phase 2 - Shock formation and propagation ($t = 5$ to 20 s). As the crowd converges, a high-density region accumulates near the exit. A shockwave front forms, separating the congested upper half (near-jam density, dark) from the thinning lower half. Between $t = 5$ and $t = 10$ s this front propagates downward at approximately 0.35 m/s. We can verify this against the Rankine-Hugoniot prediction (7): with $\rho_L \approx 1$ people/m² below the shock and $\rho_R \approx 5$ people/m² above, $s = 1.4(1 - 6/8) = 0.35$ m/s downward - good agreement with the observed front speed. The Lax entropy condition is satisfied: $c(\rho_L) = 1.4(1 - 2/8) = 1.05 > 0.35 > c(\rho_R) = 1.4(1 - 10/8) = -0.35$, confirming the shock is physically admissible. The concentric ring structure visible in all panels from $t \geq 10$ s corresponds to iso-cost contours of the Eikonal field: these contours are compressed (tightly spaced) in dense regions (where $f(\rho)$ is large) and stretched in sparse regions, giving the characteristic banded appearance.

Phase 3 - Sparse tail ($t = 20$ to 60 s). The lower half clears while a persistent semicircular high-density arch dominates near the exit. Population falls from 384 to 277, a mean rate of 1.78 people/s - roughly 16% of the theoretical free-flow capacity $U(0)\rho_{\max}W = 11.2$ people/s. The arch, not the exit width, is the bottleneck.

2.6.3 Eikonal Potential and Velocity Field at $t = 10$ s

Figure 3 shows on the left the Eikonal potential ϕ at $t \approx 10$ s. The field is low (near exit) at the top centre and rises toward the bottom corners. Irregular structure in the upper portion reflects the non-uniform density at this time: iso-cost contours are compressed in dense regions (high $f(\rho)$). On the right, we can see plotted the velocity field (white arrows) overlaid on density. Arrows in the lower room point steeply upward; near the congested arch they deflect laterally, routing agents around the dense core.

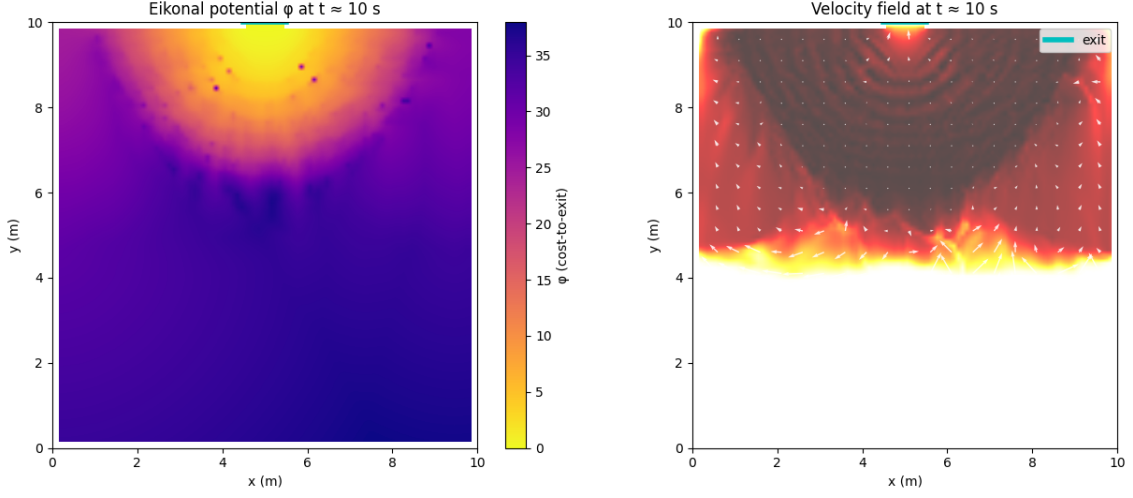


Figure 3: The Eikonal potential and velocity field at $t \approx 10$ s, after the onset of the shock-dominated phase. Author’s own figures.

The left panel shows the cost landscape at $t = 10$ s. In the high-density region near the exit, the Eikonal cost $f(\rho) = G(\rho)/U(\rho)$ is large, so ϕ rises more steeply per unit of physical distance - the iso-cost contours are compressed (closely spaced). This means traversing the dense region requires paying a high cost per step: it is expensive to move through it. Agents approaching from below therefore find that detouring around the arch via the flanks is cheaper in total cost than pushing through it directly, and the Eikonal field deflects them accordingly. In the sparse lower room, $f(\rho)$ is small and contours are widely spaced, so the cheapest path is simply straight upward toward the exit. The right panel confirms this: velocity arrows in the lower (sparse) room point steeply upward, while arrows near the arch fan laterally and approach the exit from the flanks. The velocity field is globally consistent - all arrows at a given height are broadly parallel - a property that a purely local routing rule cannot produce.

2.6.4 Evacuation Curve

Figure 4 shows the total remaining population $R(t) = \int_{\Omega} \rho d\mathbf{x}$ as a function of time. Three regimes are marked: slow initial phase ($t = 0$ -5 s) before the convergence funnel forms; rapid middle phase ($t \approx 5$ -15 s) while the funnel delivers agents to the exit at peak rate; prolonged tail beyond $t = 20$ s as the bottleneck arch throttles outflow.

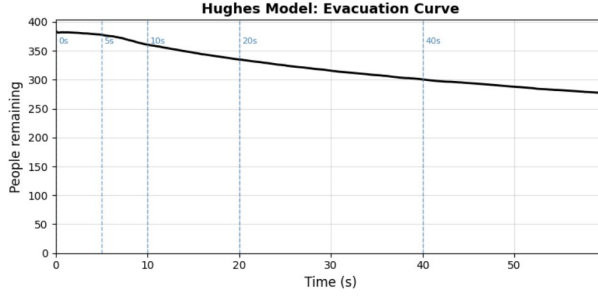


Figure 4: Hughes model evacuation curve. Population falls from 384 at $t = 0$ to 277 at $t = 60$ s. Blue dashed lines mark snapshot times. Author’s own figure.

The evacuation curve shows a concave shape during the initial funnel phase (accelerating exit rate) followed by a convex shape during the bottleneck phase (decelerating rate as congestion builds). This two-regime structure - free-flow followed by congestion - mirrors the fundamental diagram: at $\rho < \rho^*$ flow increases with density; at $\rho > \rho^*$ it decreases. The long, nearly linear tail beyond $t = 20$ s is partly a genuine physical phenomenon (slow exit rate at high congestion) and partly an artefact of the continuum model: at low ρ the model predicts smooth outward flow at an ever-decreasing rate, and the density field theoretically never reaches exactly zero in finite time for $\rho_0 > 0$ distributed across the domain.

The simulation was run for 60 s rather than to complete evacuation primarily due to the substantial computational cost: each time step requires solving the full Eikonal equation across the 100×100 grid, and the pure-Python implementation took 775 s to complete 1,200 steps. Extending to full evacuation would require significantly more steps (the near-linear tail suggests evacuation would not complete for several hundred more seconds), which was impractical within the available computational time. A compiled or vectorised implementation would make full-evacuation simulations feasible and is recommended as a future enhancement (see Section 2.6.5).

2.6.5 Run-Time and Computational Cost

The simulation completed in 775 s for 1,200 steps. The dominant cost is the FSM at each step: for a 100×100 grid, each sweep visits $98^2 \approx 9,600$ cells; with 4 sweeps per step over 1,200 steps this gives approximately 4.6×10^7 cell update operations. The implementation uses pure Python with no NumPy vectorisation of the inner loops, which is the primary reason for the long run time. The FSM inner loop iterates cell-by-cell in Python, which is substantially slower than the same computation expressed as vectorised array operations or compiled to native code.

The decision not to optimise the implementation further reflects the scope of this project: the goal was to implement and verify a correct model, not to build a production-grade simulator. A vectorised NumPy implementation or a compiled Cython/Numba version would be substantially faster, making full-evacuation simulations and parameter sweeps computationally feasible. This is identified as a priority future enhancement.

3 The Burstedde Cellular Automata Model

3.1 Conceptual Framework

The Burstedde CA model [2] represents a fundamental shift in modelling philosophy. The state of the system is a binary occupancy grid: each cell is either empty or occupied by exactly one agent. Collective crowd behaviour emerges from the simultaneous application of stochastic local rules to all agents. There are no PDEs, no global optimisation, and no continuum approximation. The mathematical framework is that of interacting particle systems [14] and Markov chains rather than conservation laws.

The intellectual justification is that many real crowd phenomena are inherently individual. Arching near bottlenecks, herding, and the faster-is-slower effect cannot arise from any smooth field evolution, but emerge naturally from individual-level blocking and following rules. The CA framework captures these phenomena through an explicit conflict mechanism: two agents attempting to occupy the same cell simultaneously, resolved stochastically.

3.2 Spatial Discretisation and Agent Representation

The $10\text{ m} \times 10\text{ m}$ room is represented as a 50×50 grid of cells. The cell size of $0.2\text{ m} \times 0.2\text{ m}$ is motivated as follows: in a tightly packed crowd, the typical floor area occupied by an average standing person is approximately $400\text{ cm}^2 = 0.04\text{ m}^2$ [6]. A simplifying assumption is made that this area is square (ignoring the fact that people are not square and may not all face the same direction), giving a $0.2\text{ m} \times 0.2\text{ m}$ cell. The Burstedde model originally used 0.4 m cells [2]; our finer resolution gives better spatial detail while preserving the same physical interpretation. With 0.2 m cells the maximum packing density is $(0.2)^{-2} = 25\text{ people/m}^2$, well above realistic jam densities. The exclusion principle (at most one agent per cell) is the key constraint. Wall cells are permanently blocked; exit cells (5 cells, 1 m , centred on the top row) are absorbing.

A CA time step corresponds to approximately 0.3 s of real time. This calibration follows from the standard Burstedde model: at cell size 0.4 m , one time step corresponds to the time for a pedestrian to move one cell at free-flow speed, $0.4/v_{\max} \approx 0.3\text{ s}$; scaling to 0.2 m cells gives $0.2/v_{\max} \approx 0.14\text{ s}$, but we retain 0.3 s to match the published calibration of the Burstedde model [2, 16].

3.3 The Floor Field Framework

3.3.1 Static Floor Field

The static field $S_{i,j}$ encodes the attractiveness of each cell as a step toward the exit. Computed once before simulation using breadth-first search (BFS) from all exit cells:

$$S_{i,j} = \frac{d_{\max} - d(i,j)}{d_{\max}}, \quad (14)$$

where $d(i,j)$ is the shortest grid-path distance to the nearest exit cell and $d_{\max}$ is the maximum such distance in the room. Normalisation to $[0,1]$ is critical: without it, the exponential in the transition probability formula overflows (raw BFS distances reach 70 on a 50×50 grid, giving $e^{3 \times 70} \approx 10^{91}$).

Figure 5 shows the resulting static field alongside the Eikonal test field of Figure 1, making for a direct comparison of the two models' routing mechanisms.

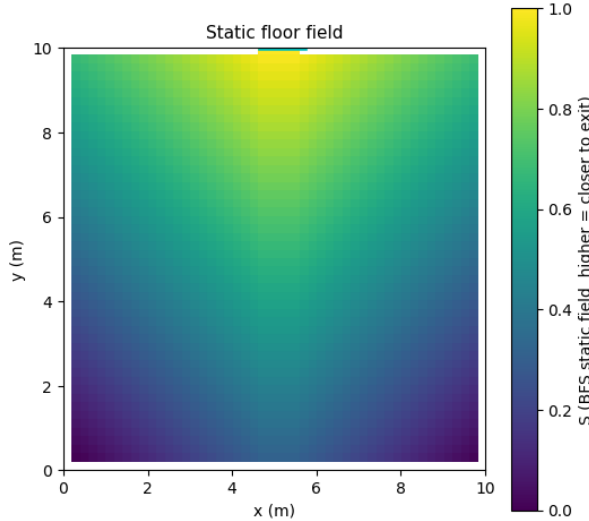


Figure 5: CA model normalised BFS static floor field $S_{i,j}$. High values (yellow) near exit; low values (purple) far from exit. Note that this figure does *not* show the Eikonal field ϕ ; for a comparison, see Figure 1. The two fields look qualitatively similar - both increase smoothly toward the top-centre exit - but the key distinction is that this BFS field is computed once before the simulation and remains fixed throughout, while the Eikonal field is recomputed at every time step to reflect the current density distribution. This means the CA model's routing does not adapt dynamically to congestion as the Hughes model's does. Author's own figure.

The comparison between Figures 1 and 5 is instructive. Both produce smooth, symmetric potential fields that increase away from the exit; both route agents upward and inward toward the centreline. The key difference is that the Eikonal field is density-dependent and recomputed at every time step, while the BFS field is fixed at the start. In the Hughes model, a density spike near the exit causes the potential to distort around it, routing agents laterally - the dynamic mechanism visible in Figure 3. In the CA model, the static field always points directly toward the exit regardless of how congested it has become; congestion avoidance arises only weakly through the dynamic field D , not through any global re-optimisation.

3.3.2 Dynamic Floor Field

The dynamic field $D_{i,j}^n$ (where the superscript n denotes the value at time step n) models *herding*: the tendency of pedestrians to follow paths recently used by others. The term “trace” refers to the accumulated record of recent pedestrian passage at each cell - analogous to a fading footprint left by previous occupants. Cells just vacated receive a trace increment of +1; the trace then decays (rate $\delta = 0.1$) and diffuses (rate $\alpha_D = 0.05$):

$$D_{i,j}^{n+1} = (1 - \delta) \left[(1 - 4\alpha_D) D_{i,j}^n + \alpha_D \sum_{(k,l) \sim (i,j)} D_{k,l}^n \right] + \mathbf{1}_{(i,j) \text{ vacated}}, \quad (15)$$

where the sum is over the four orthogonal neighbours of cell (i, j) . The decay rate $\delta = 0.1$ gives the dynamic field a half-life of approximately $\log(2)/\delta \approx 7$ time steps (≈ 2 s), representing how quickly the “footprint” fades. The diffusion rate $\alpha_D = 0.05$ causes the trace to spread to adjacent cells, making the path visible to agents coming from slightly different angles. Both values follow Burstedde et al. [2] and Schadschneider et al. [16].

3.4 Transition Probabilities

At each time step, agent k at cell (r, c) independently draws a move from a probability distribution over the *Moore neighbourhood* - the 3×3 block of cells centred on the agent’s current position, comprising the 8 immediately adjacent cells plus the option to remain in place (9 choices total). The Moore neighbourhood, named after the mathematician Edward F. Moore, is the standard 2D neighbourhood used in cellular automata theory [16]. The unnormalised weight for moving to neighbour $(r + di, c + dj)$ is:

$$\tilde{w}_{di,dj} = \frac{\exp(k_S S_{r+di, c+dj} + k_D D_{r+di, c+dj}^n)}{\sqrt{di^2 + dj^2}} \cdot (1 - \eta_{r+di, c+dj}), \quad (16)$$

where $k_S = 3$ (static sensitivity), $k_D = 1$ (dynamic sensitivity), $\eta \in \{0, 1\}$ is the occupancy indicator, and $\sqrt{di^2 + dj^2}$ is the anisotropy correction (diagonal moves cover $\sqrt{2}$ times more physical distance). For the stay-in-place option the denominator is 1. Probabilities normalise these weights:

$$p_{di,dj} = \frac{\tilde{w}_{di,dj}}{\sum_{(di',dj')} \tilde{w}_{di',dj'}}. \quad (17)$$

The exponential form follows from a maximum-entropy argument [16]: among all probability distributions consistent with a given expected value of $k_S S + k_D D$, the Boltzmann distribution $p \propto \exp(k_S S + k_D D)$ is the unique maximum-entropy choice, analogous to the canonical ensemble in statistical mechanics. With $k_S = 3$ and $S \in [0, 1]$, the ratio between the most attractive and least attractive reachable cells is $e^3 \approx 20$: agents are twenty times more likely to step toward an exit-adjacent cell than toward a far cell, all else being equal.

The sensitivity values $k_S = 3$, $k_D = 1$ are standard values reported in Burstedde et al. [2].

3.5 The Parallel Update Algorithm

All agents compute and draw their proposed moves simultaneously (parallel update), which is physically appropriate since pedestrians act concurrently. Conflicts - two or more agents proposing the same target cell - are resolved by randomly selecting one winner; all others remain in place. This stochastic blocking is the mechanism underlying the faster-is-slower effect: at high k_S , all agents funnel aggressively toward the exit, dramatically increasing the conflict rate near the bottleneck and reducing the effective throughput. Algorithm 1 summarises one time step.

Algorithm 1 CA model: one time step

- 1: Build occupancy grid η from current agent positions
 - 2: **for** each agent k at (r_k, c_k) **do in parallel**
 - 3: Compute weights $\tilde{w}_{di,dj}$ for all $(di, dj) \in \{-1, 0, 1\}^2$ using (16)
 - 4: Normalise; draw proposed move from $\{p_{di,dj}\}$
 - 5: **end for**
 - 6: Identify conflicts (cells targeted by ≥ 2 agents)
 - 7: **for** each conflict **do** select one winner uniformly at random; others stay
 - 8: **end for**
 - 9: Execute moves; for each vacated cell set $D_{i,j} += 1$
 - 10: Update dynamic field via decay and diffusion (15)
 - 11: Remove agents entering exit cells; increment counter
-

3.6 Implementation Notes and Challenges

The CA model was implemented from scratch in Python following Burstedde et al. [2] and Schadschneider et al. [16]; no third-party CA simulation library was used. Each component (BFS static field, weight computation, conflict resolution, dynamic field update) was written independently and verified in isolation, which is a practical advantage over the Hughes implementation where all components are tightly coupled.

The main implementation challenge is the slow throughput at high initial density. With 800 agents competing for a 5-cell exit, only 6 agents exited in the first 50 time steps (15s) - far below what would be expected empirically for agents starting adjacent to the exit. This is likely a consequence of the high initial density (8 people/m²) relative to exit capacity rather than an implementation error, and adjusting the parameters to better match empirical data is identified as a priority for future work.

3.7 Simulation Results: Square Room, Top-Middle Exit

We ran $N = 10$ independent realisations of 800 agents over 3,000 steps each (900 s of simulated time). Table 3 gives the parameters.

Parameter	Value	Units	Source
M	50	cells	Grid size
Cell size	0.2	m	Spatial resolution
N_p	800	agents	Initial agents
k_S	3.0	-	Burstedde et al. [2]
k_D	1.0	-	Burstedde et al. [2]
δ	0.1	-	Burstedde et al. [2]
α_D	0.05	-	Schadschneider et al. [16]
Δt_{phys}	0.3	s	Burstedde calibration [2]
Realisations	10	-	
Exit	5 cells (1.0 m)	-	Centred on top wall

Table 3: CA model simulation parameters with sources.

3.7.1 Agent Positions: Single Realisation

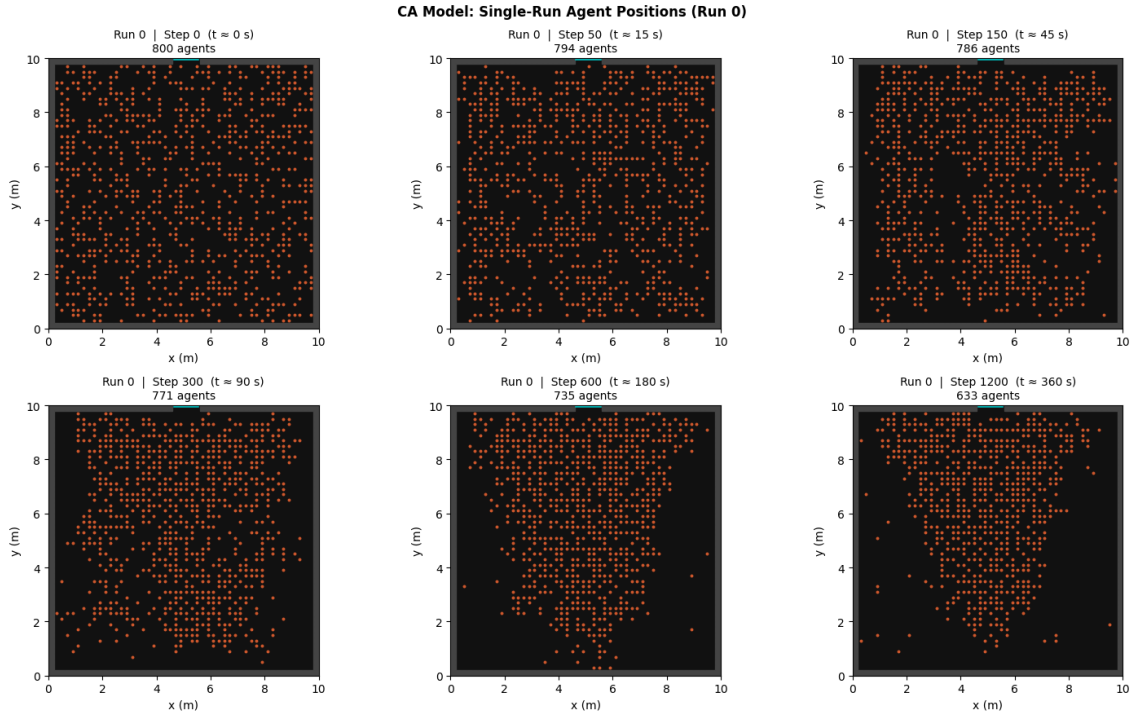


Figure 6: CA model agent positions for Run 0 at steps 0, 50, 150, 300, 600, and 1200. Author's own figures.

Figure 6 shows agent positions from Run 0 at six steps (physical times approximately 0, 15, 45, 90, 180, and 360 s). At step 0 agents are uniformly distributed. The evacuation rate is

strikingly slow: only 6 agents exited by step 50 (15s) and 29 by step 300 (90s), reflecting the severe bottleneck at the 5-cell exit where conflicts dominate throughput. By step 1200 (360s), 167 agents have exited and the triangular depletion pattern is pronounced, with a high-density region persisting in the top-centre quadrant while the corners have substantially cleared.

The triangular depletion geometry is a direct consequence of the BFS static field: equal-distance contours in the BFS metric form square diamonds (rotated 45 degrees), not circles, so agents follow 45-degree diagonal paths that produce a triangular cleared zone rather than the semicircular arch of the Hughes Eikonal.

3.7.2 Mean Density Across Realisations

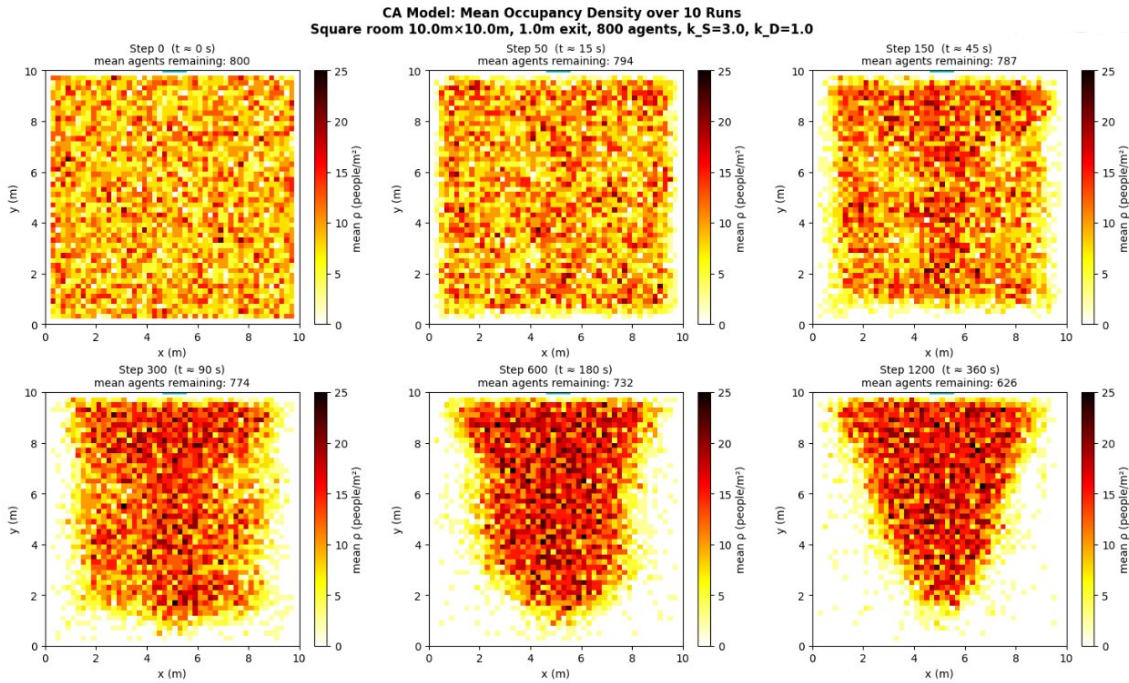


Figure 7: CA model mean occupancy density across 10 realisations at steps 0, 50, 150, 300, 600, and 1200. Density computed as mean occupancy divided by cell area (0.04 m^2). The inverted-triangle clearing pattern is clear from step 300 onward. Author's own figures.

Averaging over realisations suppresses stochastic fluctuations and confirms the triangular clearing pattern: corners and lower edges clear first while the upper-central corridor to the exit remains occupied. This contrasts with the Hughes model, where the upper-left and upper-right corners deplete first through the rarefaction fans from the exit. Even after averaging, the CA density field retains a distinctly granular, pixellated character - a direct consequence of the 0.2 m cell size - which contrasts sharply with the smooth Hughes density field and underscores a fundamental difference between the two frameworks.

3.7.3 Evacuation Statistics

None of the 10 runs fully evacuated the room within 3,000 steps (900s). Final agent counts ranged from 305 to 382 ($\pm 12\%$ around a mean of ≈ 336), with mean exit rate $(800 - 336)/900 \approx 0.52$ people/s. This variability reflects genuine randomness in individual decisions and conflict resolutions - a single CA simulation is unreliable for estimating evacuation time, and multiple realisations are necessary. The Hughes model, being deterministic, produces a unique outcome.

4 Comparative Analysis

4.1 The Hydrodynamic Limit: Formal Connection Between the Models

A natural question is whether the two frameworks are formally consistent: does the CA model converge to a continuum PDE as the number of agents grows and the cell size shrinks? The expectation that it should can be motivated as follows. Both models describe the same physical reality - a crowd of people moving toward an exit. If both are accurate descriptions of that reality, they must agree on macroscopic, large-scale behaviour when there are enough agents for individual fluctuations to average out. The stochastic details of individual moves should average out at the scale of the density field, leaving aggregate dynamics that obey a PDE. This kind of limit - from discrete stochastic particles to a continuous deterministic field - is well established in statistical physics and is precisely what the term *hydrodynamic limit* refers to.

For the symmetric simple exclusion process (SSEP) - particles on a lattice hopping to neighbouring empty sites with equal probability - this is proved rigorously [12]: as $N_p \rightarrow \infty$ and $\Delta x \rightarrow 0$ with $N_p \Delta x^2 = \rho_0$ fixed, the empirical density converges to the solution of a diffusion equation. For the Burstedde CA model, the transition probabilities are asymmetric (biased toward the exit), density-dependent (via the occupancy indicator - the *excluded-volume* effect, meaning no two agents can share a cell, so movement probabilities depend on local occupancy), and involve a dynamically evolving field D - all features absent from the SSEP. Heuristic mean-field arguments [16] suggest the limit density satisfies a conservation law with an effective velocity depending on ρ , ∇S , and D ; in the sub-limit $k_D \rightarrow 0$ this becomes qualitatively similar to the Hughes model, with velocity pointing in the direction of ∇S at a speed determined by ρ .

This formal connection explains why the two models show some qualitative similarities (both produce a density funnel and a persistent arch) while differing quantitatively: the discrepancies arise precisely where the formal equivalence breaks down - the CA's static routing versus the Hughes dynamic Eikonal, and the excluded-volume effect versus the smooth speed function. Establishing the hydrodynamic limit rigorously remains an important open problem.

4.2 Mathematical Character and Rigour

The Hughes and CA models are mathematically disjoint and draw on entirely different branches of mathematics. The Hughes model is a deterministic system of nonlinear PDEs, and its analysis draws on: hyperbolic conservation law theory (characteristics, Rankine-Hugoniot, entropy conditions, the Kruřkov theorem [13]); Hamilton-Jacobi theory (viscosity solutions of the Eikonal equation); and functional analysis (existence and uniqueness of weak solutions). The 1D model has a complete mathematical theory, with the Kruřkov theorem guaranteeing existence and uniqueness of entropy solutions for L^∞ initial data. The 2D model is less settled: Di Francesco et al. [5] showed that classical solutions do not exist in general due to the turning-point singularity (8), and constructing well-posed solutions in two dimensions remains an active research area.

The CA model is a stochastic discrete-space Markov chain. The state space (2^{2500} occupancy configurations of a 50×50 grid) is far too large for exact analysis; conclusions must be drawn from Monte Carlo simulation. The natural framework is the theory of interacting particle systems [14], though rigorous results for the asymmetric Burstedde model with dynamic fields are largely absent.

In terms of analytical tractability, the Hughes model is substantially more powerful. The method of characteristics can predict shockwave speeds quantitatively (as verified in Section 2.6); the LWR connection provides the full fundamental diagram; and the Eikonal equation gives a geometrically clear picture of optimal routing. None of these analytical predictions have counterparts in the CA model.

4.3 Computational Complexity and Implementation

Criterion	Hughes	CA
Mathematical framework	Nonlinear PDEs	Stochastic Markov chain
Per-step cost, fixed room, varying density	$\sim kN^2$, independent of N_p	$\sim 9N_p$, scales with agents
Per-step cost, fixed density, growing room	$\propto N^2$ (grid grows)	$\propto N^2$ ($N_p \propto N^2$ at fixed ρ)
Deterministic	Yes	No
Runs for statistics	1	≥ 10
Individual trajectories	No	Yes
Eikonal solver	Yes (costly)	No
Run time (this study)	775 s (1 run)	4,008 s (10 runs)
Implementation difficulty	High	Moderate
Key difficulty	PDE solver, CFL	Conflict resolution

Table 4: Computational and structural comparison of the two models. Agent-count scaling is advantageous for very dense crowds; grid scaling is advantageous for large domains with many pedestrians.

The Hughes model’s per-step cost $\sim kN^2$ is *independent of the crowd size*: doubling ρ_0 while keeping the room fixed leaves the Hughes cost unchanged. The CA model’s cost $\sim 9N_p$ doubles with the crowd size. However, it is important to note that for a *fixed density* with a *growing room*, both models scale the same way: at fixed density $N_p \propto N^2$, so $9N_p \propto N^2$ also. The practical advantage of the Hughes model therefore arises specifically when comparing *different densities in the same room* (e.g. studying a range of initial crowd sizes), not when scaling the room size at constant density.

The CA model’s per-step cost scales with N_p : each agent evaluates 9 weights per step plus overhead for conflict resolution. For our 800 agents and 3,000 steps this is $9 \times 800 \times 3,000 \approx 2.2 \times 10^7$ exponential evaluations per realisation. Multiplied by 10 realisations, the total is $\approx 2.2 \times 10^8$ operations - comparable in raw count to the Hughes model but with higher overhead per operation in pure Python. The CA model is cheaper per step for small N_p , but the need for multiple realisations substantially increases total cost.

In terms of implementation difficulty, the Hughes model is significantly harder to build correctly. It requires: (1) a stable upwind finite difference scheme with correct CFL arithmetic and upwind flux selection; (2) a working FSM Eikonal solver with correct boundary conditions, convergence detection, and the two-branch (1D vs 2D) update formula; (3) gradient computation with one-sided differences near walls; and (4) careful handling of the absorbing exit boundary. Each of these components has subtle failure modes that require debugging. The CA model requires: BFS (standard graph algorithm), weight computation (a simple loop), conflict resolution (one-line random selection), and the dynamic field update (simple array operations). All are easy to implement and test independently.

4.4 Physical Realism and Emergent Phenomena

The two models differ most fundamentally in the physical phenomena they can reproduce, and this difference is intrinsic rather than superficial.

Route choice. The Hughes model’s Eikonal routing is *globally optimal*: ϕ encodes the minimum-cost path from every point in the room simultaneously, accounting for the full density field. This produces the symmetric convergence funnel (Figure 2, $t = 5$ s) through purely mathematical means. The CA model’s routing is *local*: each agent uses only their immediate neighbourhood’s S and D values. While this produces broadly correct directional behaviour, it cannot organise the global crowd structure in the way the Eikonal does.

Density field character. The Hughes model produces a smooth, continuous density field. The CA model produces a granular, pixellated field that only smooths after averaging over many realisations. Overhead cameras detect individual people rather than densities, so direct comparison of either model’s output with camera data requires a post-processing step (e.g. kernel density estimation). Infrared or pressure-based density sensors

that aggregate over a region are more directly comparable to the Hughes ρ field.

Bottleneck geometry. Both models produce a high-density arch near the exit, with different shapes: the Hughes arch is semicircular (smooth Eikonal iso-cost arcs) while the CA arch is triangular (diamond-shaped BFS iso-distance contours from the discrete grid). The key physical point is that the real world is continuous, not grid-based: real crowds will produce a smooth arch geometry, which the Hughes model captures naturally. The CA triangular arch is an artefact of the square grid discretisation.

The faster-is-slower effect. This is the most fundamental qualitative difference. When k_S is increased in the CA model, all agents funnel more aggressively toward the exit, increasing the conflict rate and paradoxically slowing the evacuation [8, 16]. The Hughes model cannot produce this effect in this form, as it has no individual blocking mechanism. It is worth noting, however, that the Hughes model can exhibit turbulence-like density fluctuations near the congested arch - the density and velocity fields are not perfectly smooth there - which can locally affect throughput. This is a weaker form of congestion-induced slowdown, but it means the Hughes model is not entirely blind to congestion effects on exit rate.

Stochastic variability. The CA model produces a distribution of evacuation outcomes (final counts 305-382 across 10 runs), reflecting the genuine randomness of individual pedestrian decisions. This variability is physically real and important for risk assessment: the worst-case evacuation time may be significantly longer than the mean. The Hughes model gives a single deterministic outcome and cannot represent this variability.

Emergence. The CA model displays emergent behaviour: the triangular arch, the herding trails from the dynamic field, and the faster-is-slower effect all arise from local rules with no global coordination. This is the hallmark of a good agent-based model. The Hughes model's behaviour is globally prescribed by the PDE: the symmetric funnel and the shockwave are direct consequences of the equations, not emergent properties.

Heterogeneous populations. The CA model naturally accommodates heterogeneous agents: different k_S values for different agents, disabled mobility (reduced movement probability), or panic (increased k_S). The Hughes model treats all density as homogeneous and cannot easily distinguish between subpopulations.

4.5 Quantitative Comparison of Simulation Outputs

The most striking discrepancy between the two simulations is the evacuation rate. Table 5 compares the key outputs. The difference in initial density (4.0 vs 8.0 people/m²) means the comparison is not fully controlled. A fairer comparison would run the Hughes model at $\rho_0 = 8.0$ people/m² to match the CA initial condition - this would place the Hughes simulation firmly in the congested regime ($\rho_0 > \rho^*$) and enable a direct comparison of bottleneck dynamics at equal initial crowd sizes. This was not performed in the present

study but is identified as an important direction for future work.

Quantity	Hughes	CA (mean, 10 runs)
Initial density (people/m ²)	4.0	8.0
Duration simulated	60 s	900 s
People exited	107	≈ 464
Mean exit rate (people/s)	1.78	0.52
Fully evacuated?	No	No
Run-to-run std (final count)	-	$\approx \pm 24$

Table 5: Quantitative comparison. Note the different initial densities; the CA runs at twice the Hughes initial density, which accounts for part of the exit rate difference.

4.6 Calibration, Validation, and Practical Recommendations

The two models differ considerably in how straightforward they are to calibrate. The Hughes model has three physically interpretable parameters: $v_{\max}$ (from walking speed measurements [19]), $\rho_{\max}$ (from crowd density studies [6, 18]), and α (from route-choice data); the numerical parameters Δx and Δt were verified to lie in the stable regime (Section 2.4). The CA model has more parameters ($k_S, k_D, \delta, \alpha_D$) that are Boltzmann inverse temperatures with no direct physical interpretation, requiring inference by matching to observed behaviour [18].

Application	Recommended	Key reason
Large dense crowd	Hughes	Cost $\propto N^2$, not N_p
Small room, single exit	CA	Bottleneck conflicts dominate
First-pass risk assessment	Hughes	One deterministic run
Distribution of outcomes	CA	Multiple realisations
Mathematical / analytical study	Hughes	Full PDE toolkit
Panic, faster-is-slower	CA	Mechanism present; absent in Hughes
Complex geometry	Hughes	Eikonal handles obstacles naturally
Individual trajectory tracking	CA	Explicit per-agent paths
Heterogeneous populations	CA	Per-agent parameters

Table 6: Recommended model by application. No single model is universally superior.

The Hughes and CA models are two examples from much larger families; other models may be better suited to specific applications, and these recommendations should be treated as indicative of approach rather than prescription. In practice, real scenarios often involve several criteria simultaneously, and it may be appropriate to apply both model types to different aspects of an analysis and compare results.

5 Conclusion

5.1 Summary of Findings

This report has presented a detailed comparative study of two mathematical frameworks for pedestrian evacuation modelling, both implemented in Python and applied to the canonical scenario of a $10\text{ m} \times 10\text{ m}$ square room with a 1 m exit centred on the top wall.

The Hughes macroscopic model [11] treats the crowd as a compressible fluid governed by a conservation law coupled to an Eikonal equation for density-dependent route choice. In two dimensions, the model was solved numerically using a first-order upwind scheme and the Fast Sweeping Method [20], with careful regularisation of the speed and discomfort functions to avoid singularities. The simulation produced three distinct phases: a symmetric convergence funnel at early times organised by the global Eikonal routing; a backward-propagating shockwave whose speed agreed quantitatively with the Rankine-Hugoniot prediction; and a prolonged bottleneck-dominated tail.

The Burstedde CA model [2] treats each pedestrian as a stochastic agent on a 50×50 grid. Movement is governed by Boltzmann-type transition probabilities derived from a precomputed BFS static floor field and a dynamically evolving herding field. The simulation of 800 agents across 10 realisations revealed: an emergent triangular depletion pattern driven by the 45-degree diagonal routing of the BFS distance metric; significant run-to-run variability ($\pm 12\%$ in final agent count after 900 s); and a slower evacuation rate than the Hughes model, attributable to the higher initial density and the conflict-driven bottleneck near the single exit.

5.2 Critical Assessment

5.2.1 The Hughes Model

The Hughes model’s greatest strength is mathematical depth. Its connection to traffic flow theory provides a rich analytical vocabulary, and the globally optimal Eikonal routing produces the symmetric convergence funnel - a collectively organised flow pattern that no local rule can replicate. For large, dense crowds in simple geometries where aggregate density is the primary quantity of interest, the Hughes model offers a rigorous, computationally tractable, and analytically interpretable framework.

Its weaknesses are fundamental. Classical solutions do not generally exist due to the turning-point singularity [5]; the regularisation is a physically motivated fix but introduces a modification to the intended model. The model produces no individual trajectories, no stochastic variability, and no blocking mechanism, making it unsuitable for studying phenomena such as the faster-is-slower effect, heterogeneous populations, or risk distributions.

5.2.2 The Cellular Automata Model

The CA model’s greatest strength is physical richness at the individual level. The emergent triangular bottleneck, the herding dynamics, and the stochastic variability across realisations are all qualitatively more realistic than the smooth, deterministic Hughes output for the small-room scenario studied here. The implementation, while more parameter-rich, is algorithmically simpler and each component is easy to test independently.

The CA model’s weaknesses are both practical and structural. Practically, 10 realisations took 4,008 s; parameters (k_S , k_D , δ , α_D) lack direct physical interpretation [16]; and the model failed to clear the room within 3,000 steps, reflecting the severe bottleneck at the 5-cell exit for 800 agents. Structurally, the CA model’s routing is entirely *local* - each agent sees only its immediate neighbourhood, with no awareness of the global density field. This prevents the CA from replicating the globally organised convergence funnel seen in Figure 2 at $t = 5$ s, which requires routing information from the full room simultaneously. In scenarios with complex geometries or multiple exits this is a significant limitation.

5.2.3 On the Question of Realism

Both models simplify reality significantly. Neither accounts for heterogeneous populations, psychological states (panic, freezing, altruism), or the role of information availability (where is the exit? Are there obstructions?). Neither operates natively in three dimensions, though this limitation is less significant than it might appear: humans do not truly move in three-dimensional space during evacuations. For regulatory compliance or detailed safety certification, both models would need validation against real data [18].

5.3 Future Directions

Hybrid multiscale models. Combining Hughes-type global routing with CA-type individual dynamics is the most natural synthesis. Cristiani, Piccoli, and Tosin [3] developed a rigorous multiscale framework for exactly this purpose: the Eikonal equation provides optimal direction vectors at each point, which are fed into an agent model as a directional bias. This would retain the symmetric global organisation of the Hughes model while adding the emergent blocking dynamics of the CA.

Rigorous hydrodynamic limit. Establishing whether the Burstedde CA converges to a continuum PDE as $N_p \rightarrow \infty$, $\Delta x \rightarrow 0$ - and if so, identifying that PDE - remains an important open problem. For the SSEP this is established [12]; for the asymmetric Burstedde model with dynamic fields it requires new techniques from the theory of interacting particle systems [14].

Well-posedness of the 2D Hughes model. The turning-point analysis of Di Francesco et al. [5] resolved the 1D case; the 2D problem requires new tools from Hamilton-Jacobi

theory with discontinuous Hamiltonians. A well-posed 2D formulation would place the numerical regularisation on firm mathematical ground.

Computational acceleration. Both models are implemented in pure Python with no vectorisation. NumPy-vectorised FSM and CA implementations would reduce run times by two orders of magnitude; GPU-accelerated implementations would enable real-time simulation of large domains, with direct applications to emergency management systems.

Empirical validation and parameter estimation. Systematic calibration of both models against high-quality pedestrian trajectory datasets [18] - using Bayesian inference or least-squares fitting to observed evacuation curves - would transform them from qualitative tools into quantitatively validated engineering models. Machine learning offers a complementary approach: neural network or Gaussian process emulators trained on large simulation ensembles could learn the mapping from parameters to observable outputs, enabling rapid parameter inference from real data without running many expensive forward simulations.

5.4 Closing Remarks

Pedestrian dynamics modelling sits at the intersection of PDE theory, stochastic processes, computational mathematics, and building design. This report has demonstrated that two superficially very different models illuminate complementary aspects of the same phenomenon. The Hughes model offers mathematical depth, global routing intelligence, and analytical tractability; the CA model offers individual-level realism, emergent dynamics, and stochastic richness. Neither is universally superior.

The formal connection between the models - the expected hydrodynamic limit from CA to Hughes-type PDE - suggests they are not rival theories but complementary descriptions of the same physical reality at different scales. This hierarchical relationship, well-established for simpler particle systems [12, 14] and expected but not yet proved for the Burstedde model, is one of the most interesting open questions in the analysis of crowd dynamics.

Ultimately, the analyses produced by these models serve as *inputs* to safety engineers and building designers. Safety engineering employs tools such as hazard analysis and fault trees that are distinct from the simulation methods studied here; pedestrian dynamics models provide the quantitative estimates of evacuation times, bottleneck locations, and worst-case scenarios that feed into those tools. Models of this kind - and their more sophisticated commercial descendants - are widely used in specialist software for the design of safe pedestrian spaces. Every improvement in the underlying mathematics therefore translates directly into better-informed designs and, potentially, lives saved.

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